

Submission PHYS1-MID-SUB03

Question PHYS1-MID-Q01

started with a familiar rule, but the key condition is wrong:

$$v = v_{\text{initial}} + at = 4 + 3(5) = 19 \text{ m/s}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q02

started with a familiar rule, but the key condition is wrong:

$$N = mg = 58.8 \text{ N}; \text{ friction} = \mu N = 11.8 \text{ N}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q03

started with a familiar rule, but the key condition is wrong:

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 8)}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q04

started with a familiar rule, but the key condition is wrong:

$$p_{\text{total, before}} = p_{\text{total, after}}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q05

started with a familiar rule, but the key condition is wrong:

$$\text{Impulse is area under } F(t): J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}.$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q06

started with a familiar rule, but the key condition is wrong:

$$v = v_{\text{initial}} + at = 7 + 3(2) = 13 \text{ m/s}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q07

started with a familiar rule, but the key condition is wrong:

$$N = mg = 19.6 \text{ N}; \text{ friction} = \mu N = 3.9 \text{ N}$$

I treated variables/constraints as independent, so this conclusion is not justified.

Question PHYS1-MID-Q08

started with a familiar rule, but the key condition is wrong:

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 9)}$$

I treated variables/constraints as independent, so this conclusion is not justified.